Project 3 — Tasks

Optimal Risk-Averse Contracting Strategy for a Portfolio of Renewables

ECON-138

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Instructions

Submission: Through Gradescope. Check the assignment page for the deadline, file format, and any updated logistics. **Note:** two groups will give their final presentation on Wednesday of the second week (instead of the Thursday class).

Planning your presentation timing: Plan for a 30-minute presentation in total. Allocate about 60% of the time ($\approx$ 18 minutes) to Tasks 1–3 and about 40% ($\approx$ 12 minutes) to Task 4. The split is indicative — if your group finishes Tasks 1–3 in less time, spend the remainder on Task 4 rather than stretching the first part.

Grading: Composite across the three class meetings, from Thursday of the first week through Thursday of the second week. The oral defense is central to the grade: the written report and the code matter, but what we evaluate above all is each group member's ability to engage with the material live and explain the choices behind the work. A clean report that the group struggles to discuss in real time will carry less weight than a more modest report whose authors clearly understand it.

AI-assistance policy: You are welcome to use AI tools (Claude, ChatGPT, Copilot, and the like) in preparing this project. The goal of this section is not to police those tools but to ensure that they support your learning rather than substitute for it. Three guidelines apply:

(1) *The oral defense carries the grade.* The submitted artifacts (report, code, figures) are evaluated through the conversation we have during the presentation. If the work is genuinely yours, you will be in a comfortable position to discuss any part of it.

(2) *Individual questions during the presentation.* During each group's presentation we may ask any one of you to walk through a particular block of code or analysis in your own words, to run a small perturbation requested on the spot (for example: "set $P = 70$, predict what happens to the CVaR curve before re-running, then run it"), and to connect the result back to the theory. The intent is conversational — we are trying to understand how each of you engaged with the project — so prepare to speak about any part of the work, not only the part you wrote.

(3) *AI-use note at the end of each task.* Close each task with a short note (at most one slide) describing where AI assisted the work, what you asked it, what you kept and what you set aside, and how you checked the output before including it. Disclosing AI assistance is the expected norm here, and it is part of becoming a thoughtful user of these tools; this practice also makes the conversation in (2) more productive for everyone.

At our discretion during the presentation, we may also invite one of you to sketch a derivation by hand on the board — for example, the CVaR LP of Task 1. Treat this as an opportunity to show fluency with the formulation rather than a test, and prepare for it the same way you prepare for a class discussion.

As with all coursework at Stanford, the Honor Code applies; please keep your AI use transparent so we can focus on the substance of your work.

Deliverables:

(i) the **presentation report** (PDF), containing the intermediate results discussed in class and the items requested on Gradescope, including a short AI-use note at the end of each task;

(ii) the **Julia code** that produced those results.

Software: Julia + JuMP + HiGHS. Gurobi is optional for the largest runs.

Use of the source spreadsheet. All tasks below must be accomplished on the basis of the group's own code. The spreadsheet `Risk-Averse Optimal Renewable Portfolio - Multicontract - 2026.xlsm` **must not be used to present results by any means**: no figure, number, table, or chart in your report may be a screenshot, export, or transcription of cells from the workbook. The workbook serves *only* as a supporting reference — a fully worked example that students may inspect to understand the problem, and against which they may check whether their own results are sensible. Anything reported as a result must be produced by the group's Julia code and reproducible by re-running it.

Case-study setup and source data

Configuration. The case study takes the perspective of a Trading Company (TC) operating in the Brazilian market. The portfolio comprises three renewable units sited in two price zones:

- Biomass (Bio) and small run-of-river hydro (SH) in the *Southeast* (SE) zone;
- Wind (Wind) in the *Northeast* (NE) zone.

The TC's only consumer is a free consumer also in the NE zone, willing to enter a one-year forward contract at a fixed price P^{sell} for a volume Q^{sell} in avgMW. The whole operation is therefore the following question: *how much should the TC contract from each generator* $i \in \mathcal{I} = \{\mathsf{Bio}, \mathsf{SH}, \mathsf{Wind}\}$ *— under a chosen format (tolling or pay-as-produced) — to back the forward sale of* Q^{sell} *avgMW to the* NE *consumer?* All tasks below are built on top of this question.

Source file. All scenario data, decision variables, and benchmark results live in the spreadsheet

`Risk-Averse Optimal Renewable Portfolio - Multicontract - 2026.xlsm`.

This is the **source file** for the project: students pull the generation and spot-price scenarios from it, and use it as a **reference to debug and test their own portfolios**. The `Main` sheet is a fully working tolling/PAP/forward calculator — setting the four contract quantities on the control panel produces the annual risk metrics and the full monthly profit grid below. Any Julia implementation of the tasks must reproduce these numbers exactly for the same panel inputs.

Data format: months × scenarios. Throughout the workbook, scenario data is laid out as a months × scenarios grid:

- *Rows* index the month of the year, $t \in T = \{\text{Jan}, \ldots, \text{Dec}\}$, $|T| = 12$ — one operating point per month;
- *Columns* index the scenario, $\omega \in \Omega = \{1, \ldots, 2000\}$.

> This is the only structural difference with respect to the theory document: the theory uses a generic period index t, typically read as hourly; here each t aggregates a full calendar month. The number of hours in month t, H_t, is tabulated on the `Main` sheet (744, 672, 744, 720, 744, 720, 744, 744, 720, 744, 720, 744; total 8760 h). The per-month profit booked on `Main` is the per-hour profit of the theory multiplied by H_t, and the annual profit per scenario is the sum across the twelve months.

Generation Scaling Factors. For each generator i the workbook stores the Generation Scaling Factor $\mathrm{GSF}^i_{t,\omega} = g^i_{t,\omega}/\mathrm{EC}^i$ directly, in a dedicated simulation sheet:

- Wind GSF on `WindSimulation`, with $\mathrm{EC}^{\mathsf{Wind}} = 11.41$ avgMW;
- Small Hydro GSF on `SmallHydroSimulation`, with $\mathrm{EC}^{\mathsf{SH}} = 17.50$ avgMW;
- Biomass GSF on `BiomassSimulation`, with $\mathrm{EC}^{\mathsf{Bio}} = 17.59$ avgMW. Biomass is deterministic and seasonal — $\mathrm{GSF}^{\mathsf{Bio}}_{t,\omega} = 0$ in Jan–Apr and Dec, $\mathrm{GSF}^{\mathsf{Bio}}_{t,\omega} = 1.706$ in May–Nov (bagasse from the sugarcane harvest), identical across all 2000 columns.

The generation in avgMW per (month, scenario) at full capacity, $g^i_{t,\omega} = \mathrm{GSF}^i_{t,\omega}\,\mathrm{EC}^i$, is also tabulated on each simulation sheet for visual inspection.

Spot prices. Both zones are jointly simulated on the `SpotPriceSimulation` sheet: one block for the Southeast (SE) zone and one block for the Northeast (NE) zone. Under the configuration of this case study,

- the forward sale and the wind tolling/PAP flows clear at the Northeast price $\pi^{\mathsf{NE}}_{t,\omega}$ (consumer and wind plant both in NE);
- the small hydro and biomass tolling/PAP flows clear at the Southeast price $\pi^{\mathsf{SE}}_{t,\omega}$.

Where the benchmark outputs live. Use the `Main` sheet to test your code:

- monthly profit per scenario, $\Pi_{t,\omega}$ [\$];
- annual profit per scenario, $\Pi_\omega = \sum_{t \in T} \Pi_{t,\omega}$;
- annual risk metrics: $\mathbb{E}[\Pi]$, Median, $\mathrm{CVaR}_{95\%}$, $\mathrm{VaR}_{95\%}$, Risk, and the combined metric $\rho_{\alpha,\lambda}$.

Every task in Parts 1–5 below is solved against this same dataset: the GSF blocks and the two spot-price blocks are the only stochastic inputs — everything else (contract prices P^i, P^{sell}, hours H_t, capacities EC^i, λ, α) is a parameter set on the panel.

1 Task 1 — PQ-Risk illustration and CVaR linear program

In this task you will (i) implement the Conditional Value-at-Risk functional as the sampled Rockafellar–Uryasev linear program of equation (19) of the theory document, and (ii) use that function to reproduce the single-period illustration of Section 2.4 of the theory — including its Figure 1 — and then enrich the plot with two additional curves (risk premium and standard deviation).

(a) Implement the CVaR function. Write a Julia function

```
cvar(profit, alpha)
```

that takes a vector of N equiprobable profit realizations $(\Pi_1, \dots, \Pi_N)$ and a tail level $\alpha \in (0, 1]$, and returns $\widehat{\mathrm{CVaR}}_\alpha(\Pi)$ as the optimal value of the linear program in equation (19) of the theory:

$$\widehat{\mathrm{CVaR}}_\alpha(\Pi) \;=\; \max_{z,\,\{\delta_\omega\}_{\omega=1}^N} \quad z \;-\; \frac{1}{\alpha N}\sum_{\omega=1}^{N}\delta_\omega \quad \text{s.t.} \quad \delta_\omega \;\geq\; z - \Pi_\omega, \;\; \delta_\omega \;\geq\; 0, \;\; \forall\, \omega = 1, \dots, N.$$

Use `JuMP` + `HiGHS`. **Include the code of this function in your report.**

(b) Reproduce Figure 1 of Section 2.4. Sweep Q over a fine grid in $[0, 15]$. For each Q, build the two-scenario profit vector $\big(\Pi(\omega_1; Q),\ \Pi(\omega_2; Q)\big)$, call `cvar(profit, 0.05)`, and store the returned value as $\widehat{\mathrm{CVaR}}_{0.05}(Q)$. Plot, on the same axes as in Figure 1 of Section 2.4:

- the two scenario lines $\Pi(\omega_1; Q)$ and $\Pi(\omega_2; Q)$;
- the expected-profit curve $\mathbb{E}[\Pi](Q) = 250$;
- the CVaR curve $\widehat{\mathrm{CVaR}}_{0.05}(Q)$ produced by your function, placed exactly where the worst-case envelope sits in Figure 1 of Section 2.4. Since $\alpha N = 0.05 \times 2 = 0.10 < 1$, the LP returns the worst-of-two profit at each Q, so this curve must coincide with the dashed red envelope of that figure.

(c) Add the risk premium and the standard deviation. On the same plot add:

- the *risk premium* $\mathrm{RP}(Q) \;:=\; \mathbb{E}[\Pi](Q) - \widehat{\mathrm{CVaR}}_{0.05}(Q)$;
- the *standard deviation* $\sigma_\Pi(Q) \;:=\; \sqrt{\mathrm{Var}\big[\Pi(Q)\big]}$ of the random profit.

Both curves are well-defined functions of Q; report them in the same units ($) as the profit curves.

(d) Comment on the results. Be ready to explain the following in your presentation:

(i) why the expected-profit curve is flat in Q on this instance, and what it tells you about the limitations of an expected-profit-only criterion for choosing Q;

(ii) how the CVaR curve, the risk-premium curve, and the standard-deviation curve all reach their best level (highest CVaR; lowest RP; lowest σ_Π) at the same $Q^\star = 5$, and how this $Q^\star$ relates to $\mathbb{E}[g] = 10$ — i.e., why a risk-averse trader contracts *less* than the expected generation;

(iii) what a $\rho_{\alpha,\lambda}$-maximizing trader would choose for $\lambda = 0$, $\lambda = 0.5$, and $\lambda = 1$, and why the optimum collapses to a single point $Q^\star = 5$ as soon as $\lambda > 0$;

(iv) which spot-price scenarios compose the CVaR at $Q = 0$, $Q = 5$, and $Q = 10$. Identify, in each case, the scenario(s) that fall in the lower tail picked up by the LP — low-price (sunny) or high-price (cloudy) — and explain how the choice of Q changes which state becomes the "bad" state of the contracted position.

(e) Repeat (a)–(d) with $P = 55$. Redo all four items above for the same two-scenario primitive but with the contract price raised from $P = 50$ to $P = 55$ (all other data unchanged). In particular: rebuild the two scenario profit lines, regenerate the plot of (b) including the new $\widehat{\mathrm{CVaR}}_{0.05}(Q)$ curve, recompute the risk-premium and standard-deviation curves of (c), and re-answer the four commentary items of (d). Highlight what changes relative to the $P = 50$ case — in particular, that $\mathbb{E}[\Pi](Q)$ is no longer flat in Q (since $P > \mathbb{E}[\pi] = 50$ creates a forward risk premium that the trader earns on every unit contracted), and how this tilts the trade-off between expected profit and CVaR, shifting the $\rho_{\alpha,\lambda}$-optimal $Q^\star$ as λ moves between 0 and 1.

2 Task 2 — Wind-only tolling on the case-study data

This task moves from the two-scenario illustration of Task 1 to the case-study data of the workbook. The TC contracts only wind through a tolling contract, with all other generators switched off, and varies Q^{sell} to study the joint behavior of expected profit, CVaR, and the certainty-equivalent metric on real scenarios.

(a) Sweep Q^{sell} and replicate slides 40 and 41. Adopt the configuration of slide 40 of the case-study deck (`Week 9 -- Market structures and Electricity Contract Auctions in Latin America`, Case 1, $\lambda = 0.5$): the TC buys only wind under a tolling contract, with the other parameters fixed at

$$P^{\mathrm{Sell}} = 140 \text{ \$/MWh}, \quad P^{\mathsf{Wind}} = 100 \text{ \$/MWh}, \quad Q^{\mathsf{Wind}} = 11.41 \text{ avgMW}, \quad Q^{\mathsf{SH}} = Q^{\mathsf{Bio}} = 0,$$

and tail level $\alpha = 0.05$. Sweep Q^{sell} on the grid $\{0.00, 0.01, 0.02, \ldots, 15.00\}$ avgMW. For each value, build the annual-profit vector $(\Pi_1(Q^{\mathrm{sell}}), \ldots, \Pi_{2000}(Q^{\mathrm{sell}}))$ by aggregating the per-month profits across the twelve months of the horizon (using the hours H_t from the workbook), and compute three annual metrics:

- the expected value $\mathbb{E}\big[\Pi(Q^{\mathrm{sell}})\big]$;
- the CVaR $\widehat{\mathrm{CVaR}}_{0.05}\big[\Pi(Q^{\mathrm{sell}})\big]$, using the function written in Task 1;
- the certainty-equivalent metric $\rho_{\alpha,\lambda}\big[\Pi(Q^{\mathrm{sell}})\big] = \lambda\,\widehat{\mathrm{CVaR}}_{0.05}\big[\Pi(Q^{\mathrm{sell}})\big] + (1-\lambda)\,\mathbb{E}\big[\Pi(Q^{\mathrm{sell}})\big]$.

Plot the three curves against Q^{sell} on the same axes. Locate the maximizer $Q^{\mathrm{sell}\star}$ of $\rho_{\alpha,\lambda}$ on the grid and check that the corresponding triple $\big(\mathbb{E}\big[\Pi(Q^{\mathrm{sell}\star})\big], \widehat{\mathrm{CVaR}}_{0.05}\big[\Pi(Q^{\mathrm{sell}\star})\big], \rho_{\alpha,\lambda}\big[\Pi(Q^{\mathrm{sell}\star})\big]\big)$ approximately matches the values reported on slide 40 ($Q^{\mathrm{sell}\star} \approx 9.88$ avgMW). Repeat the entire sweep for $\lambda = 0.99$ and verify that the result approximately matches slide 41 ($Q^{\mathrm{sell}\star} \approx 9.12$ avgMW).

(b) Which spot-price scenarios drive the tail? Still in the wind-only tolling configuration of (a), produce a single figure that overlays the Northeast spot-price time-series of the 2000 scenarios on one set of axes:

- plot all 2000 scenarios as time-series paths in light gray — one continuous line per scenario, connecting $\pi^{\mathsf{NE}}_{1,\omega}, \pi^{\mathsf{NE}}_{2,\omega}, \ldots, \pi^{\mathsf{NE}}_{12,\omega}$ across the twelve months $t \in T$. The figure therefore contains 2000 paths, one for each scenario ω. Use lines only, with no markers, for a cleaner background;

- in **red**, plot — on top of the gray paths, so they remain clearly visible — the time-series paths of the scenarios that compose $\widehat{\mathrm{CVaR}}_{0.05}$ when $Q^{\mathrm{sell}} = 15$ (i.e., the 100 worst-profit scenarios at that decision);

- in **blue**, plot — also on top of the gray background — the time-series paths of the scenarios that compose $\widehat{\mathrm{CVaR}}_{0.05}$ when $Q^{\mathrm{sell}} = 0$ (the 100 worst-profit scenarios at that decision).

Before producing the plot, be ready to explain in your presentation *what you expect* to see. Base the prediction on the structure of the per-period profit $\Pi_{t,\omega}(Q^{\mathrm{sell}})$ — the profit booked in month t, scenario ω, as a function of the forward-sale decision Q^{sell}. Writing it first as the sum of the forward leg and the wind tolling leg, and then rearranging to put the spot price in evidence:

$$\begin{aligned}
\Pi_{t,\omega}(Q^{\mathrm{sell}}) \;&=\; (P^{\mathrm{Sell}} - \pi^{\mathsf{NE}}_{t,\omega})Q^{\mathrm{sell}} \;+\; \pi^{\mathsf{NE}}_{t,\omega}\,\mathrm{GSF}^{\mathsf{Wind}}_{t,\omega}\,Q^{\mathsf{Wind}} \;-\; P^{\mathsf{Wind}}\,Q^{\mathsf{Wind}} \\
&=\; P^{\mathrm{Sell}}\,Q^{\mathrm{sell}} \;-\; P^{\mathsf{Wind}}\,Q^{\mathsf{Wind}} \;+\; \pi^{\mathsf{NE}}_{t,\omega}\,(\mathrm{GSF}^{\mathsf{Wind}}_{t,\omega}\,Q^{\mathsf{Wind}} \;-\; Q^{\mathrm{sell}}).
\end{aligned}$$

In the second line, the first two terms are deterministic and the third is the only stochastic component: $\pi^{\mathsf{NE}}_{t,\omega}$ multiplied by the trader's net spot exposure $\left(\mathrm{GSF}^{\mathsf{Wind}}_{t,\omega}\,Q^{\mathsf{Wind}} - Q^{\mathrm{sell}}\right)$. The sign of this net exposure flips between $Q^{\mathrm{sell}} = 0$ and $Q^{\mathrm{sell}} = 15$ (long the spot vs. short the spot), and should drive your prediction of which side of the π^{NE} distribution composes the tail in each case. Then comment on what you actually observe, and back the comment with whatever summary statistics you find useful to make the point. One clean presentation is a small table, grouped by population (*red* 100, *blue* 100, and *all* 2000), reporting at least: the mean annual-average π^{NE}, its 5% and 95% empirical quantiles, and the mean annual $\mathrm{GSF}^{\mathsf{Wind}}$. The pattern should make it visible how the same trader, with the same physical asset, can have opposite "bad states" depending on the size of the forward sale.

3 Task 3 — Willingness-to-supply curve and the value of the portfolio

This task moves from a fixed forward-sale volume to the *offer curve* of the trading company: how much it would be willing to sell forward at each contract price P^{sell}, given a maximum demand from the consumer and a fixed risk preference. The exercise then compares the curve obtained by jointly optimizing the three generators with the curves obtained by optimizing each generator separately, to make the value of the portfolio visible both in volume and in certainty equivalent.

Setup. Choose one supply-side contract format — either *tolling* or *pay-as-produced* — and apply it to all three generators throughout the task. State your choice and the reason for it in one sentence. The other parameters are fixed at the slide-50 configuration:

$$P^{\mathsf{SH}} = P^{\mathsf{Wind}} = P^{\mathsf{Bio}} = 100\ \$/\mathrm{MWh}, \quad \lambda = 0.5, \quad \alpha = 0.05, \quad Q^{\mathrm{sell}}_{\max} = 100\ \mathrm{avgMW},$$

together with the per-asset capacity ceilings $\mathrm{EC}^{\mathsf{SH}} = 17.50$, $\mathrm{EC}^{\mathsf{Wind}} = 11.41$, $\mathrm{EC}^{\mathsf{Bio}} = 17.59$ avgMW. Sweep P^{sell} on a fine grid in $[0, 250]$ \$/MWh (e.g. step 5 \$/MWh).

At each value of P^{sell} on the grid, solve four versions of the contracting LP of the adjusted version (considering all the specificities discussed in this case study) of model (27) of the theory document:

- three *single-asset* runs — one for each $i \in \{\mathsf{Wind}, \mathsf{SH}, \mathsf{Bio}\}$ — with the other two generators forced to $Q^j = 0$;
- one *joint-portfolio* run with all three generators available.

For each run record the optimal forward sale $Q^{\text{sell}\star}(P^{\text{sell}})$, the total acquired energy certificate $\sum_i Q^{i,\star}(P^{\text{sell}})$, the expected profit $\mathbb{E}[\Pi(\mathbf{q}^\star)]$, the CVaR $\widehat{\mathrm{CVaR}}_{0.05}[\Pi(\mathbf{q}^\star)]$, and the certainty equivalent $\rho_{\alpha,\lambda}[\Pi(\mathbf{q}^\star)]$.

(a) Willingness-to-supply plot (slide-50 layout). On one figure with P^{sell} on the horizontal axis, plot the eight curves of the four runs:

- the optimal forward sale $Q^{\text{sell}\star}(P^{\text{sell}})$ for each of the four configurations (solid lines, one color per configuration);
- the corresponding total acquired EC, $\sum_i Q^{i,\star}(P^{\text{sell}})$, for each configuration (dashed lines, same color as the solid line).

The qualitative shape should match slide 50 of the case-study deck: each curve stays at zero up to a threshold value of P^{sell} (below the contract price $P^i = 100$ the trader has no incentive to sell), then rises and saturates at the asset's EC ceiling.

(b) Certainty equivalent and risk premium: portfolio vs. sum of individuals. On a second figure with P^{sell} on the horizontal axis, plot:

- the portfolio certainty equivalent $\rho_{\alpha,\lambda}[\Pi(\mathbf{q}^\star_{\text{portfolio}}(P^{\text{sell}}))]$ and the sum of the three single-asset certainty equivalents $\sum_i \rho_{\alpha,\lambda}[\Pi(\mathbf{q}^\star_i(P^{\text{sell}}))]$, as solid lines;
- the portfolio *risk premium* $\mathrm{RP}_{\text{portfolio}}(P^{\text{sell}}) := \mathbb{E}[\Pi(\mathbf{q}^\star_{\text{portfolio}}(P^{\text{sell}}))] - \widehat{\mathrm{CVaR}}_{0.05}[\Pi(\mathbf{q}^\star_{\text{portfolio}}(P^{\text{sell}}))]$ and the sum of the three single-asset risk premia $\sum_i \mathrm{RP}_i(P^{\text{sell}})$, as dashed lines.

Using the properties of the certainty equivalent described in the theory document, demonstrate and comment on the relation between the certainty equivalent of the joint optimal portfolio and the sum of the certainty equivalents of the individual portfolios.

Risk–return frontier of the joint portfolio. As a complementary view, fix $P^{\text{sell}} = 140$ \$/MWh and the joint-portfolio configuration, and sweep the risk-aversion weight λ on the grid $\{0.200, 0.205, 0.210, \ldots, 1.000\}$ (step 0.005, i.e. 161 points). For each λ solve the joint LP and record two quantities of the resulting random profit:

- *return*: $\mathbb{E}[\Pi(\mathbf{q}^\star_\lambda)]$;
- *risk*: $\mathbb{E}[\Pi(\mathbf{q}^\star_\lambda)] - \widehat{\mathrm{CVaR}}_{0.05}[\Pi(\mathbf{q}^\star_\lambda)]$.

Plot a scatter with risk on the horizontal axis and return on the vertical axis; each point is one value of λ, and the curve traced as λ moves from 0.200 (less risk-averse) to 1.000 (fully risk-averse on the CVaR tail) is the trader's risk–return frontier under the certainty-equivalent metric. Comment briefly on the shape of the frontier and identify the endpoints in terms of the trader's preferences.

(c) Snapshot at $P^{\text{sell}} = 140$ \$/MWh. Fix the contract price at $P^{\text{sell}} = 140$ \$/MWh and inspect the four optimal solutions in detail: report $Q^{\text{sell}\star}$ and the contracted quantities $Q^{\mathsf{Wind}\star}, Q^{\mathsf{SH}\star}, Q^{\mathsf{Bio}\star}$ (zero where the corresponding asset is not in the run), together with $\mathbb{E}[\Pi^\star]$, $\widehat{\text{CVaR}}_{0.05}[\Pi^\star]$, and $\rho_{\alpha,\lambda}[\Pi^\star]$, for each of the three single-asset configurations and for the joint portfolio. Lay the four solutions side by side in a small table.

Then comment specifically on the presence of small hydro in the *joint-portfolio* solution: why is $Q^{\mathsf{SH}\star} > 0$ at the joint optimum even though small hydro on its own (the SH-only run) decides not to contract at all at this price? Use the seasonality of the relevant quantiles — e.g., the 5%, 50%, and 95% month-by-month quantiles — of GSF^{SH} and of the spot prices π^{SE}, π^{NE} to make your point. You may also find it useful to plot the empirical density (across the 2000 scenarios) of the per-scenario temporal correlation between the spot price and the corresponding GSF — $\text{Corr}_t(\pi^{\mathsf{SE}}_{t,\omega}, \text{GSF}^{\mathsf{SH}}_{t,\omega})$ for each ω, and analogously $\text{Corr}_t(\pi^{\mathsf{NE}}_{t,\omega}, \text{GSF}^{\mathsf{Wind}}_{t,\omega})$. Use these statistics to explain what role small hydro plays inside the portfolio that it cannot play on its own, and tie the answer back to the certainty-equivalent gap discussed in (b).

4 Task 4 — Your project

Build on the concepts of Tasks 1–3 — risk assessment, optimization, and portfolio thinking applied to power markets — and develop them into a self-contained piece of original analysis. You may keep working with the workbook data of this case study, or bring in data from a different power system (e.g. ERCOT, CAISO, Nord Pool, the Iberian MIBEL, the Brazilian ONS public data) if the question you want to ask is better posed there. The aim is to showcase *one interesting effect* or *one relevant application* that this framework expresses and the data can answer.

Directions to seed your thinking. You are not expected to follow any single one of these; they only sketch the territory that fits the project:

- *Portfolio extensions.* A different cohort of generators (battery + solar + wind + hydro); cross-zone operation with a limited transmission link; cross-market arbitrage; long/short positions on assets the trader does not own physically.
- *Finer time resolution.* Move from monthly aggregates to typical daily profiles of generation and spot price (e.g. representative weekday and weekend daily shapes, possibly distinct across seasons) and show what the monthly aggregation hides — the within-day correlation between price spikes and the PV/wind shape, the value of flexibility, the role of intraday hedging.
- *Other instruments and flexibility assets.* Add a call, a put, a collar, a swap, or a swing option; introduce *demand response* on the consumer side as a flexible short position; or *add a reservoir upstream the small hydro* to turn the run-of-river output into a (partially) dispatchable resource. Price the new instrument or asset by the certainty-equivalent indifference approach used in the theory document, and compute the optimal mix with the forward sale and the tolling/PAP positions.

Deliverables for this task.

(i) a clear presentation of the question, the data, the model and what you expect;

(ii) the *model statement* — the optimization model you solve, with all notation defined and a one-line justification of each constraint;

(iii) the *numerical study* that produces the effect: at least one main figure and one supporting table;

(iv) a *discussion* (one to two slides) interpreting the result and connecting it back to the theory of risk-averse contracting;

(v) where applicable, a short *comparison* with the project baseline (Tasks 1–3) so the value of the extension is visible.

Scope note. This is the most open part of the project. It does not need to be ambitious — it needs to be clear, correct, and interesting. A small, well-executed extension is preferred over a large, half-finished one. Talk to us early in the second week if you want to validate the direction before committing.

In the presentation, talk through your optimization model: state its decision variables, its objective, and its constraints, and use the model itself to explain what you are trying to achieve and why each piece is there. The model is the bridge between the question you are asking and the numbers you will show; spend time on it.

Data quality matters as much as the model. Use realistic data — public scenarios from a recognized system operator or market, or the workbook data of this case study — and do not select an application whose data is fabricated, implausible, or stripped of the joint structure (e.g. price–generation correlations) that makes the risk question meaningful in the first place. Briefly state the source and any cleaning/processing steps in deliverable (i).